



PREMIUM

Full-Length Mock Test

Mock 03

Petroleum Engineering

GATE PE

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Reservoir Engineering — Well Testing]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.2 [1 Mark] [EASY] [Production Engineering — Artificial Lift]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.3 [1 Mark] [EASY] [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.4 [1 Mark] [EASY] [Reservoir Engineering — Material Balance]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.5 [1 Mark] [EASY] [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.6 [1 Mark] [EASY] [Drilling Engineering — Drilling Fluids]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.7 [1 Mark] [EASY] [Reservoir Engineering — PVT Analysis]

Primary reservoir drive mechanism is:

A) Gas cap

B) Water

C) Solution gas

D) Gravity

Q.8 [1 Mark] [EASY] [Production Engineering — Flow Assurance]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.9 [1 Mark] [EASY] [Drilling Engineering — Drilling Fluids]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.10 [1 Mark] [EASY] [Reservoir Engineering — Well Testing]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.11 [1 Mark] [EASY] [Production Engineering — Flow Assurance]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.12 [1 Mark] [EASY] [Drilling Engineering — Drilling Fluids]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.13 [2 Marks] [MODERATE] [Reservoir Engineering — Material Balance]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.14 [2 Marks] [MODERATE] [Production Engineering — Sand Control]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit

B) Only lubricate

C) Only seal

D) Only transport

Q.15 [2 Marks] [MODERATE] [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.16 [2 Marks] [MODERATE] [Reservoir Engineering — PVT Analysis]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.17 [2 Marks] [MODERATE] [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.18 [2 Marks] [MODERATE] [Drilling Engineering — Casing Design]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.19 [2 Marks] [MODERATE] [Reservoir Engineering — Well Testing]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.20 [2 Marks] [MODERATE] [Production Engineering — Well Completions]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.21 [2 Marks] [MODERATE] [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

- A) Gas cap

B) Water

C) Solution gas

D) Gravity

Q.22 [2 Marks] [MODERATE] [Reservoir Engineering — Reservoir Drive]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.23 [2 Marks] [MODERATE] [Production Engineering — Sand Control]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.24 [2 Marks] [MODERATE] [Drilling Engineering — Directional Drilling]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.25 [2 Marks] [MODERATE] [Reservoir Engineering — Material Balance]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.26 [2 Marks] [MODERATE] [Production Engineering — Sand Control]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.27 [2 Marks] [MODERATE] [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.28 [2 Marks] [MODERATE] [Reservoir Engineering — Well Testing]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit

B) Only lubricate

C) Only seal

D) Only transport

Q.29 [2 Marks] [MODERATE] [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.30 [2 Marks] [MODERATE] [Drilling Engineering — Directional Drilling]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.31 [2 Marks] [MODERATE] [Reservoir Engineering — PVT Analysis]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.32 [2 Marks] [MODERATE] [Production Engineering — Sand Control]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.33 [2 Marks] [HARD] [Drilling Engineering — Bit Selection]

Primary reservoir drive mechanism is:

- A) Gas cap
 - B) Water
 - C) Solution gas
 - D) Gravity
-

Q.34 [2 Marks] [HARD] [Reservoir Engineering — PVT Analysis]

Drilling mud primary functions include:

- A) Remove cuttings, cool bit
 - B) Only lubricate
 - C) Only seal
 - D) Only transport
-

Q.35 [2 Marks] [HARD] [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

- A) Gas cap

B) Water

C) Solution gas

D) Gravity

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Drilling Engineering — Directional Drilling]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.37 [2 Marks] [HARD] [Reservoir Engineering — PVT Analysis]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.38 [2 Marks] [HARD] [Production Engineering — Flow Assurance]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.39 [2 Marks] [HARD] [Drilling Engineering — Directional Drilling]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.40 [2 Marks] [HARD] [Reservoir Engineering — PVT Analysis]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Drilling Engineering — Drilling Fluids]**

Standard calculation using basic formula:

Answer: _____

Q.52 [2 Marks] [HARD] [Reservoir Engineering — Material Balance]

Standard calculation using basic formula:

Answer: _____

Q.53 [2 Marks] [HARD] [Production Engineering — Flow Assurance]

Standard calculation using basic formula:

Answer: _____

Q.54 [2 Marks] [HARD] [Drilling Engineering — Directional Drilling]

Standard calculation using basic formula:

Answer: _____

Q.55 [2 Marks] [HARD] [Reservoir Engineering — Material Balance]

Standard calculation using basic formula:

Answer: _____

Q.56 [2 Marks] [HARD] [Production Engineering — Flow Assurance]

Standard calculation using basic formula:

Answer: _____

Q.57 [2 Marks] [HARD] [Drilling Engineering — Bit Selection]

Standard calculation using basic formula:

Answer: _____

Q.58 [2 Marks] [HARD] [Reservoir Engineering — PVT Analysis]

Standard calculation using basic formula:

Answer: _____

Q.59 [2 Marks] [HARD] [Production Engineering — Well Completions]

Standard calculation using basic formula:

Answer: _____

Q.60 [2 Marks] [HARD] [Drilling Engineering — Directional Drilling]

Standard calculation using basic formula:

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	C	Q.2:	A	Q.3:	C
Q.4:	A	Q.5:	C	Q.6:	A
Q.7:	C	Q.8:	A	Q.9:	C
Q.10:	A	Q.11:	C	Q.12:	A
Q.13:	C	Q.14:	A	Q.15:	C
Q.16:	A	Q.17:	C	Q.18:	A
Q.19:	C	Q.20:	A	Q.21:	C
Q.22:	A	Q.23:	C	Q.24:	A
Q.25:	C	Q.26:	A	Q.27:	C
Q.28:	A	Q.29:	C	Q.30:	A
Q.31:	C	Q.32:	A	Q.33:	C
Q.34:	A	Q.35:	C	Q.36:	AC
Q.37:	AC	Q.38:	AC	Q.39:	AC
Q.40:	AC	Q.41:	AC	Q.42:	AC
Q.43:	AC	Q.44:	AC	Q.45:	AC
Q.46:	AC	Q.47:	AC	Q.48:	AC
Q.49:	AC	Q.50:	AC	Q.51:	10
Q.52:	10	Q.53:	10	Q.54:	10
Q.55:	10	Q.56:	10	Q.57:	10
Q.58:	10	Q.59:	10	Q.60:	10

Detailed Solutions

Question 1 [Reservoir Engineering — Well Testing]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 2 [Production Engineering — Artificial Lift]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 3 [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 4 [Reservoir Engineering — Material Balance]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 5 [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 6 [Drilling Engineering — Drilling Fluids]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 7 [Reservoir Engineering — PVT Analysis]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 8 [Production Engineering — Flow Assurance]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 9 [Drilling Engineering — Drilling Fluids]

Primary reservoir drive mechanism is:

Answer:

Solution gas drive is the most common primary mechanism.

Question 10 [Reservoir Engineering — Well Testing]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 11 [Production Engineering — Flow Assurance]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 12 [Drilling Engineering — Drilling Fluids]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 13 [Reservoir Engineering — Material Balance]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 14 [Production Engineering — Sand Control]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 15 [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 16 [Reservoir Engineering — PVT Analysis]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 17 [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 18 [Drilling Engineering — Casing Design]

Drilling mud primary functions include:

Answer:

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 19 [Reservoir Engineering — Well Testing]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 20 [Production Engineering — Well Completions]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 21 [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 22 [Reservoir Engineering — Reservoir Drive]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 23 [Production Engineering — Sand Control]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 24 [Drilling Engineering — Directional Drilling]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 25 [Reservoir Engineering — Material Balance]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 26 [Production Engineering — Sand Control]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 27 [Drilling Engineering — Directional Drilling]

Primary reservoir drive mechanism is:

Answer:

Solution gas drive is the most common primary mechanism.

Question 28 [Reservoir Engineering — Well Testing]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 29 [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 30 [Drilling Engineering — Directional Drilling]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 31 [Reservoir Engineering — PVT Analysis]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 32 [Production Engineering — Sand Control]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 33 [Drilling Engineering — Bit Selection]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 34 [Reservoir Engineering — PVT Analysis]

Drilling mud primary functions include:

Answer: (A)

Mud removes cuttings, cools bit, stabilizes well, controls pressure.

Question 35 [Production Engineering — Well Completions]

Primary reservoir drive mechanism is:

Answer: (C)

Solution gas drive is the most common primary mechanism.

Question 36 [Drilling Engineering — Directional Drilling]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 37 [Reservoir Engineering — PVT Analysis]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 [Production Engineering — Flow Assurance]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 [Drilling Engineering — Directional Drilling]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 [Reservoir Engineering — PVT Analysis]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 [Production Engineering — Sand Control]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 [Drilling Engineering — Casing Design]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 [Reservoir Engineering — PVT Analysis]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 [Production Engineering — Well Completions]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 [Drilling Engineering — Directional Drilling]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 46 [Reservoir Engineering — PVT Analysis]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 [Production Engineering — Well Completions]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 [Drilling Engineering — Directional Drilling]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 [Reservoir Engineering — Well Testing]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 [Production Engineering — Sand Control]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 [Drilling Engineering — Drilling Fluids]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 52 [Reservoir Engineering — Material Balance]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 53 [Production Engineering — Flow Assurance]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 54 [Drilling Engineering — Directional Drilling]

Standard calculation using basic formula:

Answer:

Apply the appropriate formula for the given data.

Question 55 [Reservoir Engineering — Material Balance]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 56 [Production Engineering — Flow Assurance]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 57 [Drilling Engineering — Bit Selection]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 58 [Reservoir Engineering — PVT Analysis]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 59 [Production Engineering — Well Completions]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.

Question 60 [Drilling Engineering — Directional Drilling]

Standard calculation using basic formula:

Answer: (10)

Apply the appropriate formula for the given data.
